Numerical Analysis I

Quiz 9A Name:_____

Question 1.

The number of users of an online platform has grown rapidly over several years. The table below gives the number of users (in millions) at selected times.

Year	t	Users (millions), y
2015	0	2.5
2016	1	3.8
2017	2	5.6
2018	3	8.7
2019	4	12.9
2020	5	19.8
2021	6	29.5

Fit the exponential model

$$y = c_1 e^{c_2 t}$$

to the data using model linearization.

(a) Linearize the model by taking the natural logarithm of both sides.

(b) Express the resulting problem as a linear least squares give the coefficient matrix A and the right hand side b such that we solve

$$\min_{x \in \mathbb{R}^2} \|Ax - b\|_2.$$

(c) We find $x_1 \approx 0.916$ and $x_2 \approx 0.412$. Determine the parameters c_1 and c_2 of the original exponential model.

Question 2.

Find the least squares solution of

$$\min_{x \in \mathbb{R}^2} \|Ax - b\|_2,$$

where

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \qquad b = \begin{bmatrix} 2 \\ 2 \\ 5 \\ 5 \end{bmatrix}.$$